

The theorem of the three perpendiculars

One theorem that turns every “is this angle right?” question in space into a question about a shadow.

LESSON 38–39

2 × 40 MIN

GEOMETRY 10, §16

EXTENSION

LESSON CLOCK

14'

22'

24'

22'

16'

8'

The statement	14 min
The proof	22 min
Applications	24 min
Distance to a line	22 min
Practice	16 min
Homework	8 min

LEARNING OBJECTIVES

- State the theorem of the three perpendiculars and its converse.
- Apply it to prove perpendicularity in a solid.
- Use it to find the distance from a point to a line in space.
- Recognise the three perpendiculars in a given figure.

TEXTBOOK

National	pp. 177–188
Cambridge	Core & Extended, pp. 335–338
Workbook	Exercise 16.1

01

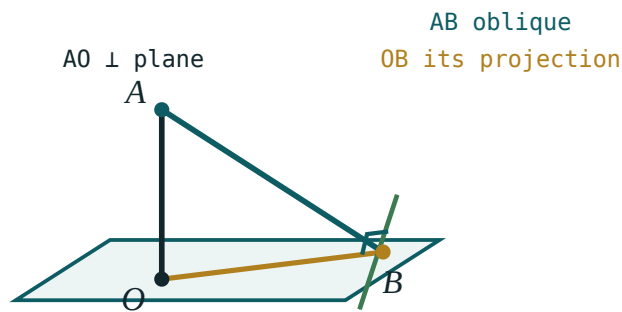
Explanation

The statement

THEOREM OF THE THREE PERPENDICULARS

Let $AO \perp \alpha$ with foot O , let AB be an oblique with projection OB , and let c be a line in α passing through B . Then

$c \perp AB \Leftrightarrow c \perp OB$



a line of the plane through B is $\perp$ AB
exactly when it is $\perp$ OB

*AO is the perpendicular, AB the oblique, OB its
projection. A line of the plane is $\perp$ to one exactly when
it is $\perp$ to the other.*

The three perpendiculars of the name are $AO \perp \alpha$, $c \perp OB$ and $c \perp AB$ — any two give the third.

The proof

Forward direction. Suppose $c \perp OB$. Since $AO \perp \alpha$ and $c \subset \alpha$, also $c \perp AO$. So c is perpendicular to two intersecting lines OB and AO of the plane AOB . By the test of Lesson 33–34, $c \perp AOB$, and therefore $c \perp AB$, since $AB \subset AOB$. ■

Converse. The same argument with AB and AO in place of OB and AO . ■

THE PATTERN TO REMEMBER

Prove perpendicularity to a **plane**, then read off perpendicularity to any line in it. Almost every space perpendicularity proof in this course has that shape.

Applications

In a cube. Is $AC_1 \perp BD$? Project AC_1 onto the base: its projection is AC . Since $BD \perp AC$ (diagonals of a square) and $BD \subset ABCD$, the theorem gives $BD \perp AC_1$ at once — no coordinates, no vectors, two lines of work.

In a pyramid. A regular square pyramid $SABCD$ with centre O . Is $SM \perp AB$, where M is the midpoint of AB ? The projection of SM is OM , which is perpendicular to AB by symmetry. So yes — and SM is the slant height.

THE LINE MUST LIE IN THE PLANE AND PASS THROUGH THE FOOT

c must be in α and meet OB at B . A line of the plane not through B is not covered by the theorem — translate it first.

The distance from a point to a line in space

To find the distance from A to a line c lying in a plane α , when $AO \perp \alpha$:

1. Drop $OB \perp c$ inside the plane — an ordinary plane construction.
2. By the theorem, $AB \perp c$ too.
3. So AB is the required distance, and $AB = \sqrt{AO^2 + OB^2}$.

Example. A is 12 above the plane; the foot O is 5 from the line c . Then the distance from A to c is $\sqrt{144 + 25} = 13$.

02 Worked examples

EXAMPLE 1 In a cube, prove $AC_1 \perp BD$.

- 1 $CC_1 \perp ABCD$, so AC is the projection of AC_1 .
- 2 $BD \perp AC$
Diagonals of a square.
- 3 $BD \subset ABCD$ and passes through the foot.
- 4 By the theorem of the three perpendiculars.

ANSWER

$$AC_1 \perp BD$$

EXAMPLE 2 A is 12 cm above a plane, and its foot O is 5 cm from a line c in the plane. Find the distance from A to c .

① Drop $OB \perp c$ in the plane; $OB = 5$.

② By the theorem, $AB \perp c$.

③ $AB = \sqrt{144 + 25} = 13$

ANSWER

13 cm

EXAMPLE 3 A regular square pyramid has base edge 10 and height 12. Find the slant height.

① $OM = 5$, the apothem of the base.

② $SO \perp ABCD$, so OM is the projection of SM .

③ $OM \perp AB$, so $SM \perp AB$.

It is the slant height.

④ $SM = \sqrt{144 + 25} = 13$

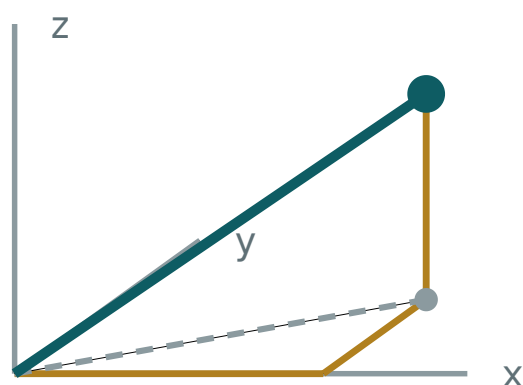
ANSWER

13

03 Interactive model

Point at the three perpendiculars on a real box before any calculation.

The three perpendiculars


 x 3

 y 2

 z 2

OM 4.12

base diagonal 3.61

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Theorem of the three perpendiculars	Uch perpendikulyar haqidagi teorema	Теорема о трёх перпендикулярах
Oblique line	Og'ma	Наклонная
Projection	Proyeksiya	Проекция
Converse theorem	Teskari teorema	Обратная теорема
Line in the plane	Tekislikdagi chiziq	Прямая в плоскости
Foot of the perpendicular	Perpendikulyar asosi	Основание перпендикуляра
Distance to a line	Chiziqqacha masofa	Расстояние до прямой
Necessary and sufficient	Zarur va yetarli	Необходимо и достаточно

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The theorem relates:

two obliques

an oblique and its projection

two planes

two normals

The line c must:

be anywhere

lie in the plane through the foot of the oblique

be perpendicular to the plane

be parallel to AO

In a cube $AC_1 \perp BD$ because:

$AC \perp BD$

$AA_1 \perp BD$

both

neither

If $AO = 12$ and $OB = 5$ then AB :

7

13

17

60

The theorem lets you replace a 3-D question with:

a 2-D question in the plane

a calculation

a guess

nothing

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. State the theorem of the three perpendiculars

ANSWER $c \perp AB \Leftrightarrow c \perp OB$

2. $AO = 3$, $OB = 4$; find AB

ANSWER 5

3. $AO = 12$, $OB = 5$; find AB

ANSWER 13

4. $AB = 25$, $OB = 24$; find AO

ANSWER 7

5. Is $AC_1 \perp BD$ in a cube?

ANSWER yes

6. The projection of AC_1 on the base

ANSWER AC

7. Where must c lie?

ANSWER in the plane, through B

Medium · the standard question

7 PROBLEMS

1. Prove $AC_1 \perp BD$ in a cube

ANSWER Three perpendiculars, with projection AC

2. A is 12 above; O is 5 from c . Distance to c

ANSWER 13

3. Regular pyramid, base 10, height 12: slant height

ANSWER 13

4. Regular pyramid, base 16, height 6: slant height

ANSWER 10

5. A is 8 above; distance to a line 15 from O

ANSWER 17

6. In a cube, is $BD_1 \perp AC$?

ANSWER yes

7. Prove $SM \perp AB$ in a regular pyramid

ANSWER $OM \perp AB$ and the theorem

Hard · stretch and reason

7 PROBLEMS

1. Cube edge a : prove $AC_1 \perp BDA_1$

ANSWER Three perpendiculars twice

2. Regular pyramid, base 12, lateral edge 10: the slant height

ANSWER 8

3. Same pyramid: the height

ANSWER $\sqrt{82} \approx 9.06$

4. A point 15 above a plane; two lines in the plane at distances 8 and 20 from O . Distances from A

ANSWER 17 and 25

5. Prove the converse of the theorem

ANSWER Same argument with AB in place of OB

6. In a cube, find the distance from A to the line B_1C_1 for edge 6

ANSWER $6\sqrt{2}$

7. Why must c pass through B ?

ANSWER Otherwise it is not in the plane AOB

07 Homework

Homework — 5 tasks

Name the perpendicular, the oblique and the projection in every solution.

1. State the theorem of the three perpendiculars and prove the forward direction.
2. In a cube, prove $BD_1 \perp AC$.
3. A is 9 cm above a plane and its foot is 12 cm from a line c . Find the distance from A to c .
4. A regular square pyramid has base edge 14 and height 24. Find the slant height and the lateral edge.

5. Explain in three sentences why the theorem reduces a space question to a plane one.